

Text Large Model + YOLOv11 + Intelligent Garbage Classification

Function Description

After the program starts, give the large model a garbage sorting command. After the large model thinks it through, it will control the robotic arm to sort the blocks on the table.

Startup

After entering the docker container, enter the following command in the docker terminal:

```
ros2 launch largemodel largemodel_control.launch.py
```

Garbage sorting has the following types of commands:

- Sort a single type of garbage: for example, sort toxic garbage
- Sort all garbage on the desktop: for example, clean up all garbage on the desktop

Sort Single Type of Garbage

Startup

After waking up the module, input by voice:

```
Clean up toxic garbage
```

Wait for the large model to think and reply. The large model will execute the `garbage_sort` function, pass in the parameter `tox`, and control the robotic arm to grip the toxic garbage label block on the desktop.

Task Planning

Call `garbage_sort('tox')` to sort toxic garbage.

Core Source Code Analysis

The `garbage_sort` function, please refer to the content in **Core Source Code Analysis** in the tutorial **【AI Model-Text Version】 - 【11.Text Large Model + YOLOv11 + Intelligent Garbage Classification】**. The voice version and text version have the same action functions; only the input method for task commands is different.

Sort All Garbage on the Desktop

Startup

After waking up the module, input by voice:

```
Clean up all garbage on the desktop
```

Wait for the large model to think and reply. The large model will execute the `garbage_sort_all()` function to sort the garbage label blocks on the desktop.

Task Planning

Call `garbage_sort_all()` function.

Core Source Code Analysis

The `garbage_sort_all()` function, please refer to the content in section 2.3 of the tutorial **【AI Model-Text Version】** - **【11.Text Large Model + YOLOv11 + Intelligent Garbage Classification】** . The voice version and text version have the same action functions; only the input method for task commands is different.